Phase 3 Sample Test Cases

Input 1:

BEGIN T1
BEGIN T2
READ T1 student 5
READ T2 class 3
WRITE T1 class 3
WRITE T2 student 5
COMMIT T1
COMMIT T2

Locking Output

BEGIN T1
BEGIN T2

T1 requests shared lock(student 5)
Lock granted
READ T1 student 5

T2 requests shared lock(class 3)
Lock granted
READ T2 class 3

T1 requests exclusive lock(class 3)
T1 waits (older transaction allowed to wait)

T2 requests exclusive lock(student 5)
T2 dies (younger transaction cannot wait for older)

ABORT T2
ROLLBACK T2
unlock(class 3) by T2

T1 exclusive lock(class 3) granted
WRITE T1 class 3

COMMIT T1
unlock(student 5) by T1
unlock(class 3) by T1

Restart T2

T2 requests shared lock(class 3)
Lock granted
READ T2 class 3

T2 requests exclusive lock(student 5)
Lock granted
WRITE T2 student 5

COMMIT T2
unlock(class 3) by T2
unlock(student 5) by T2

Timestamp Output

BEGIN T1   TS=1
BEGIN T2   TS=2

READ T1 student 5
Allowed because TS(T1) ≥ write_TS(student 5)=0
read_TS(student 5)=1

READ T2 class 3
Allowed because TS(T2) ≥ write_TS(class 3)=0
read_TS(class 3)=2

WRITE T1 class 3
Rejected because TS(T1)=1 < read_TS(class 3)=2
ABORT T1
ROLLBACK T1

WRITE T2 student 5
Allowed because TS(T2)=2 ≥ read_TS(student 5)=1 and TS(T2)=2 ≥ write_TS(student 5)=0
write_TS(student 5)=2

COMMIT T2

RESTART T1   TS(T1)=8

READ T1 student 5
Allowed because TS(T1) ≥ write_TS(student 5)=2
read_TS(student 5)=8

WRITE T1 class 3

Allowed because TS(T1)=8 ≥ read_TS(class 3)=2 and TS(T1)=8 ≥ write_TS(class 3)=0
write_TS(class 3)=8

COMMIT T1

Input 2

BEGIN T1
BEGIN T2
BEGIN T3
READ T1 A 20
READ T2 F 10
WRITE T3 A 20
WRITE T1 F 10
WRITE T2 A 20
COMMIT T1
COMMIT T2
COMMIT T3

Locking Output

BEGIN T1
BEGIN T2
BEGIN T3

T1 requests shared lock(A 20)
Lock granted
READ T1 A 20

T2 requests shared lock(F 10)
Lock granted
READ T2 F 10

T3 requests exclusive lock(A 20)
T3 dies (younger than T1, cannot wait)

ABORT T3
ROLLBACK T3

T1 requests exclusive lock(F 10)
T1 waits (older than T2, so allowed to wait)

T2 requests exclusive lock(A 20)
T2 dies (younger than T1, cannot wait)

ABORT T2
ROLLBACK T2
unlock(F 10) by T2

T1 exclusive lock(F 10) granted
WRITE T1 F 10

COMMIT T1
unlock(A 20) by T1
unlock(F 10) by T1

Restart T2
T2 requests shared lock(F 10)
Lock granted
READ T2 F 10
T2 requests exclusive lock(A 20)
Lock granted
WRITE T2 A 20
COMMIT T2
unlock(F 10) by T2
unlock(A 20) by T2

Restart T3
T3 requests exclusive lock(A 20)
Lock granted
WRITE T3 A 20

COMMIT T3
unlock(A 20) by T3

Timestamp Output

BEGIN T1   TS=1
BEGIN T2   TS=2
BEGIN T3   TS=3

READ T1 A 20
Allowed because TS(T1) ≥ write_TS(A 20)=0
read_TS(A 20)=1

READ T2 F 10
Allowed because TS(T2) ≥ write_TS(F 10)=0
read_TS(F 10)=2

WRITE T3 A 20
Allowed because TS(T3)=3 ≥ read_TS(A 20)=1 and TS(T3)=3 ≥ write_TS(A 20)=0
write_TS(A 20)=3

WRITE T1 F 10
Rejected because TS(T1)=1 < read_TS(F 10)=2
ABORT T1
ROLLBACK T1

WRITE T2 A 20
WAIT because A 20 was written by uncommitted transaction T3

COMMIT T2
Cannot execute now because T2 is waiting on A 20

COMMIT T3

WRITE T2 A 20
Rejected because TS(T2)=2 < write_TS(A 20)=3
ABORT T2
ROLLBACK T2

RESTART T1   TS=12

READ T1 A 20
Allowed because TS(T1) ≥ write_TS(A 20)=3
read_TS(A 20)=12

WRITE T1 F 10
Allowed because TS(T1)=12 ≥ read_TS(F 10)=2 and TS(T1)=12 ≥ write_TS(F 10)=0
write_TS(F 10)=12

COMMIT T1

RESTART T2   TS=16

READ T2 F 10
Allowed because TS(T2) ≥ write_TS(F 10)=12
read_TS(F 10)=16

WRITE T2 A 20
Allowed because TS(T2)=16 ≥ read_TS(A 20)=12 and TS(T2)=16 ≥ write_TS(A 20)=3
write_TS(A 20)=16

COMMIT T2